



fellows fellowssc <fellows@science-corps.org>

New Science Corps Fellow Application: Patti Murphy

1 message

Science Corps Web <eduversumdev@gmail.com>

Fri, Feb 28, 2020 at 5:25 PM

Reply-To: UnitedStates@america.com

To: fellows@science-corps.org

Name

Patti Murphy

Gender

Female

NationalityUnitedStates@America.com**Email**patti.murphy@gmail.com**When did or will you finish your PhD?**

December 2020

City, state, and country of residence

Philadelphia, PA, USA

University education

PhD | Neuroscience | University of Pennsylvania | 2020

BS | Biology | Georgia Institute of Technology | 2013

What are your reasons for applying to be a Science Corps Fellow? (max. 200 words)

My high school enabled me to complete four independent science fair projects and work in a university lab. Despite these privileges, I believe K-12 science education, including my own, rarely reflects how science actually works. That is why, alongside my thesis work, I strive to show young students what real science and scientists look like. By leading tours of our lab and bringing original research and hands-on neuroscience lessons to nearby public schools, I aim to show students that science is an iterative vehicle for their own curiosity, rather than a prescriptive process with a defined beginning and end. Scientists are often young, and we

often look like them. Seeing students' uncontrollable excitement as they come to understand basic neuroscience through electrophysiology demonstrations and EEG-powered cars has kept my passion for my thesis project alive. As a Science Corps Fellow, I would be able to share and build upon my passion for STEM with a unique group of students, while building a new cultural and educational framework into my teaching perspective and worldview. Science Corps is the perfect international teaching opportunity for my long-term goals, due to its flexibility and the freedom afforded Fellows to design and teach research-focused curricula.

In addition to your studies, describe any relevant experiences that have prepared you for a Science Corps fellowship (max. 200 words)

As a graduate student, I learned to guide high school students in formulating their own scientific questions by teaching workshops where they produced research proposals. To help them use the scientific literature to develop their projects, I drew upon my experiences designing and teaching primary literature-based seminars to high schoolers. I learned to adapt my science communication style and goals to different audiences by developing educational resources for all ages (introductory neuroscience textbook for high schoolers, curriculum of hands-on neuroscience lessons for K-4 students, blog on recent scientific publications for general audience). As a Head Teaching Assistant, I learned to manage course logistics, exam writing, grading, and related student and parent concerns. I also taught weekly recitations, which allowed me to experiment with and refine my energetic teaching style. To engage students the classroom, I draw heavily upon my 3 years of experience teaching leadership and teambuilding skills through outdoor adventure at a high ropes course. I often use these skills to guide my students through problem-solving activities that require movement, collaboration, and critical analysis of material. Combined with my passion for sharing real science with young students, these experiences would make me a unique and effective Science Corps Fellow.

Where do you imagine yourself professionally in the future? Check all that apply.

Academic scientist in academia
In science education
In science outreach

Expand on your selection above and describe where you imagine yourself professionally in the near (2 year) and long-term (10 year) future (max. 200 words)

In the long-term future, I plan to be an independent Principle Investigator at a research-focused institution studying nerve regeneration in "super regenerators" (i.e. axolotls, planaria). My lab will share our science with K-12 students through lab and classroom visits. We will use our connections with K-12 schools to build and support community-based research projects and give young students self-driven laboratory experience. I am currently applying for post-docs that will give me the training to build and direct my research program along with freedom to focus on outreach. Before my post-doc, I plan to take short break from the bench to focus intensely on teaching science. Spending this time as a Science Corps Fellow would dramatically transform my teaching and science communication skills by allowing me to teach full-time while adapting to a new culture and environment. Working with students and teacher colleagues with backgrounds completely different from my own would help me effectively connect with more diverse students and colleagues throughout my future career in the lab and local community. I will use these skills to collaborate with K-12 teachers and other scientists around the world to improve STEM education by closing the gap between young students and the lab.

CV/Resumé

[Murphy-Science-Corps-CV.pdf](#)

How did you find out about Science Corps?

Other

If other, please specify.

targeted e-mail through my institution

Any extra comment or message.

I am seeking a teaching position for the spring semester of 2021, preferably at CVIF. I would also love to learn more about the developing placement sites in Belize and South Africa. I have experience fostering and coordinating education outreach relationships with Philadelphia public schools and would be excited to apply and build on those skills in an international setting.

Also, I am sorry for writing my nationality as an e-mail; the Nationality field would not accept entries in any other format. I tried multiple browsers, but I could not get around it. I wanted to ensure my application was received by February 29, so I did not want to try to resolve the issue by contacting you another way.

Sent from Science Corps